

ENVIRONMENTAL EMERGENCIES MODULE

What to remember

Online Module Quick Reference

Each lesson of the online module is distilled to its four or five essential ideas. This document does not replace the manual, the video capsules or in-class practice. It gives you a reference point before the in-person session and a quick refresher after the course.

The online module covers the emergencies where prevention and reasoning matter more than hands-on skills: water, lightning, bites and stings, submersion. Cold-related injuries (hypothermia, frostbite) and heat-related injuries, dehydration and hyponatremia are covered in person, because they require practice in assessment and rewarming.

MODULE CONTENTS: EIGHT LESSONS

INTRODUCTION

00 Introduction to the chapter

PREVENT

01 Water disinfection

02 Lightning

BITES AND STINGS

03 Animal bites

04 Snakebites

05 Arachnid bites

06 Mosquito bites

WATER

07 Submersion incidents and rescue

00

Introduction to the chapter

Seven lessons where prevention is worth more than treatment

Why this chapter: the environment does not injure like a fall or a knife. It works over time, often silently: water that makes you sick three days later, a tick attached for 36 hours, a venom that spreads, cold water that takes away your motor control in ten minutes. Here, the right decision is made before the incident.

- 1 **Three blocks: prevent, bites and stings, water.** Drinking water and lightning are matters of preparation and reading the terrain. Bites and stings (animals, snakes, arachnids, mosquitoes) share the same logic: wound, infection, venom or transmitted disease. Submersion closes the module with a rescue rule.
- 2 **Which type of shock?** Hypovolemic for diarrhea and hemotoxic envenomations, respiratory or obstructive for submersion and some neurotoxic venoms, cardiogenic for lightning, distributive for an allergic reaction to a sting. Ask yourself the question in every lesson.
- 3 **Find out before you leave.** Animals, insects, snakes, diseases and vaccines of the region visited. What you do not know before departure, you will not be able to improvise on site. The travel clinic is part of the emergency plan.
- 4 **Rescuer safety, again.** Lightning, moving water, suspicious animal: several lessons in this module begin with a decision about your own exposure. A second patient does not save the first.
- 5 **In class: cold, heat, the body's water.** Hypothermia, frostbite, heat stroke, dehydration, hyponatremia. These emergencies are practised in person, with the hypothermia wrap and the scenarios.

LEARNER'S MANUAL

Section 10, Environmental Emergencies · Online encyclopedia

FIELD MANUAL

**Water: pages 18 and 19 ·
Environment: pages 165 to 179**

IN CLASS

Cold- and heat-related injuries, dehydration, hyponatremia

01

Water disinfection

Making water safe with simple methods suited to the field

No natural water is safe without treatment, even if clear and cold. Clean water is used for drinking and for irrigating wounds. The choice of method depends on the context, the resources and the constraints of the field; combining two techniques often gives the best result.

Method	Effective against	Limits
Boiling	Bacteria, viruses, protozoa (everything)	Fuel, time, cooling; reliable even at altitude
Filtration	Bacteria, protozoa, particles	Not viruses (too small): combine
Ultraviolet	Bacteria, viruses, protozoa	Clear water only; batteries; neither particles nor chemicals
Chemical (chlorine, iodine, chlorine dioxide)	Bacteria, viruses; protozoa depending on product and time	Contact time, cold water, turbidity, taste; iodine contraindicated for some people

Figure 1. Four methods, what they eliminate and their limits. Combining two techniques often gives the best result.

- Four complementary steps.** Clarification (removing visible particles), purification (removing some impurities without killing microbes), disinfection (destroying or inactivating pathogens: the key step), post-treatment (taste and odour). In the field, they are combined.
- Clarify first, because particles shield microbes.** Let the water settle so heavy matter sinks, filter through a cloth or a coffee filter. Cloudy water reduces the effectiveness of UV and chemicals.
- Three factors govern chemical disinfection.** Contact time (longer is more effective), temperature (cold water slows everything down), clarity. Follow the recommended dose and time, and double the time in cold water.
- Post-treatment never disinfects.** Activated carbon against odours and residues, aeration, fruit or herbs, vitamin C to neutralize the chlorine taste: only after disinfection, and only for comfort.
- Choose according to context.** Large group and fuel available: boiling. Solo hiking: filter plus chemical or UV. Muddy water: clarify, then boil or treat chemically with extended time.

02

Lightning

An unpredictable phenomenon: anticipate, protect yourself, resuscitate

Ten percent of lightning flashes reach the ground, and those are the ones that injure, directly or by spreading through the environment. Lightning follows the path of least resistance, at up to 300,000 amperes. It cannot be predicted; you read the sky and act beforehand.

- 1 **Protecting yourself when a storm approaches.** Get off the water and away from ridges, avoid isolated trees and tall objects, spread out among small trees, crouch on an insulating object (life jacket, pack), remove metal objects when possible, watch for falling trees.
- 2 **The injury that kills: cardiorespiratory arrest.** About 30% of deaths. Direct cardiac arrest, or respiratory paralysis leading to hypoxia and then secondary arrest. Immediate CPR saves lives in lightning victims: start without delay, even if several patients appear lifeless.
- 3 **Neurological injuries are the rule.** Loss of consciousness in 70% of victims, confusion, amnesia, temporary paralysis. In severe cases, brain injury and permanent sequelae. Reassess consciousness and motor function at regular intervals.
- 4 **Burns and trauma: look for the entry and exit points.** Burns are often superficial, sometimes in branching patterns (Lichtenberg figures), deeper under clothing or heated metal. Any organ located between the entry and exit points may have been affected. The shock wave can throw the patient: blunt trauma, eardrums, lungs, eyes.
- 5 **After the strike: monitor and evacuate as a precaution.** A patient who regains consciousness without apparent trauma remains at risk of secondary complications (cardiac, neurological). Precautionary medical assessment, complete PAS, spinal precautions if thrown.

03

Animal bites

A wound, an infection risk, and the question of rabies

A bite is never trivial. Irregular, deep puncture wound from a dog or cat, crushing from a large animal, possible injury to nerves, vessels and muscles. In children, bites mostly affect the face and neck.

- 1 Bleeding, then pressure irrigation.** Control the bleeding, assess the depth and the structures involved. Irrigate with drinking water using a syringe to create a jet, in quantity, to dislodge bacteria and debris without damaging the tissues.
- 2 Disinfect gently, do not close.** Diluted povidone-iodine on the wound; no harsh products. A bite is a wound at high risk of infection: sterile dressing, wound left open, monitoring for infection, tetanus status checked.
- 3 Unknown or suspicious animal: think rabies right away.** Any warm-blooded animal can carry it. A wild animal that is not afraid of humans is suspect. Never handle the animal; contact the authorities (veterinarian, animal control) if possible.
- 4 Washing with soap and water in the first hours reduces the risk of rabies.** Then mandatory medical assessment, even if the patient has received pre-exposure doses: post-exposure vaccination is given as soon as possible. Once symptoms appear, rabies is almost always fatal.
- 5 Child bitten on the face or neck: immobilize the cervical spine.** The mechanism (shaking, fall) justifies spinal precautions when there is significant trauma. Prevention: teach children not to provoke animals.

04

Snakebites

Recognize, slow the venom, evacuate

Not all bites are venomous, but when venom is injected, its effects can be serious and rapid. The only specific treatment is antivenom, in a hospital setting. Your role: slow the spread, monitor, evacuate.

- 1 Three families, two types of venom.** Vipers (rattlesnakes, vipers): mainly hemotoxic venom, pain, swelling, bleeding. Elapids (cobras, coral snakes): neurotoxic venom, weakness, vision and swallowing problems, respiratory paralysis. Colubrids: mostly harmless. Note what you saw, without chasing the animal.
- 2 Calm, lie down, immobilize below heart level.** Limit all movement of the patient and the limb. Exertion speeds the spread of venom. Remove rings and bracelets before swelling sets in.
- 3 Mark and time.** Trace the edge of the swelling on the skin with a pen every 15 minutes, noting the time. Watch for new signs: some venoms act quickly, others produce delayed symptoms.
- 4 Pressure immobilization: for neurotoxic venoms only.** Compression bandage and immobilization of the limb to slow the spread. Indicated for elapids; contraindicated for some vipers such as rattlesnakes, where it worsens local tissue damage. If unsure of the species, do not apply it.
- 5 Evacuate by carrying; never a tourniquet or suction.** Litter transport preferably; if walking is unavoidable to evacuate quickly, limit exertion to the strict minimum. Tourniquets, incision, suction and ice make things worse. Even without symptoms, medical assessment is mandatory: antivenom is species-specific and is given under monitoring.

05

Arachnid bites

Spiders, scorpions and ticks

Eight legs, no antennae, no wings. In North America, most spiders and scorpions pose no serious danger. Two spiders, one scorpion and one tick-borne disease are worth knowing.

- 1 **Brown recluse: a bite that digs in.** Light brown, violin marking, dark places, active at night. Two small marks, then a pale, painful area, sometimes necrosis and healing over several months. In children: nausea, headache, vomiting, cramps.
- 2 **Black widow: a venom that mimics a heart attack.** Glossy black, red hourglass, only the female is dangerous. Neurotoxic venom: muscle cramps, fatigue, dizziness, heavy sweating, nausea; locally, redness and mild swelling. Resolves in 48 to 72 hours, possible complications in children, older adults or immunosuppressed people.
- 3 **Scorpions: like a bee sting, with one exception.** Mainly in the North American Southwest. Local pain, redness, inflammation. The Arizona bark scorpion can cause neuromuscular effects and respiratory distress.
- 4 **Common management and prevention.** Clean and disinfect, cold for pain, hydration, analgesic, tetanus status checked, monitoring. Evacuation if systemic symptoms or allergic reaction. Children: faster and more severe progression. Shake out boots and clothing, close the tent, gloves for wood and rocks.
- 5 **Ticks and Lyme disease: remove quickly, monitor for 30 days.** Transmission after 24 to 36 hours of attachment. Fine-tipped tweezers close to the skin, slow steady pull without crushing, wash with soap and water, keep the tick, note the date and location. Early signs: fatigue, fever, body aches, erythema migrans (expanding red patch). Preventive treatment depending on region, attachment time, appearance and delay: a medical decision.

LEARNER'S MANUAL

Section 10.4, Arachnid Bites; Ticks and Lyme Disease · Online encyclopedia

FIELD MANUAL

Pages 173, 178 and 179, stings, ticks and prevention

EN LIGNE

Decision aid: preventive treatment after a tick bite

06

Mosquito bites

Transmitted diseases and insect repellents

Mosquitoes are the leading disease vectors in the world. The infections look alike at first: fever, headache, muscle pain, fatigue. Most often there is no specific treatment; prevention and prompt consultation make the difference.

Disease	Where	Key points
West Nile virus	Several continents, including North America	Flu-like, sometimes encephalitis
Malaria	Tropical areas	Any fever on return = malaria until proven otherwise; coma, organ failure
Dengue, chikungunya, Zika	Aedes mosquitoes, tropical areas	Fever, pain, rash; hemorrhagic dengue; Zika dangerous in pregnancy
Yellow fever	Africa, South America	Liver damage, hemorrhage; the vaccine is the prevention

Figure 2. The main mosquito-borne diseases.

- 1 Before departure: the travel clinic.** Vaccines, preventive medications (malaria), risks specific to the region. It is not the first aider's job to improvise prophylaxis on site.
- 2 Any fever after a tropical stay is an emergency until proven otherwise.** Symptoms may appear several days after exposure. Prompt medical consultation; in the meantime, supportive care: hydration, rest, symptom treatment.
- 3 The repellents that work.** DEET (very effective, sometimes irritating; follow the doses, especially in children, watch for dermatitis, hives, rare neurological effects) and icaridin (not before 6 months of age). Also: permethrin on clothing (toxic to animals, caution in young children), oil of lemon eucalyptus, metofluthrin. Citronella and natural products are not very effective.
- 4 Physical measures are essential.** Long, light-coloured clothing, pants tucked into socks, mosquito net, no perfume, avoid going out at night. In high-risk areas, combine everything.
- 5 Recognizing a reaction that is no longer local.** Most bites are benign. Significant swelling, fever, rash, symptoms developing over the following days: consult.

LEARNER'S MANUAL

Section 10.4, Insect-Borne Diseases;
Insect Repellents · Online
encyclopedia

FIELD MANUAL

Pages 173 and 179, stings and
prevention

LEVEL

Diseases: awareness; repellents:
knowledge

07

Submersion incidents and rescue

Analyze, decide safely, act

Drowning is respiratory impairment from immersion or submersion. It leads to lack of oxygen, respiratory arrest and then cardiac arrest. It can be fatal or non-fatal, with or without sequelae, and complications can occur hours later.

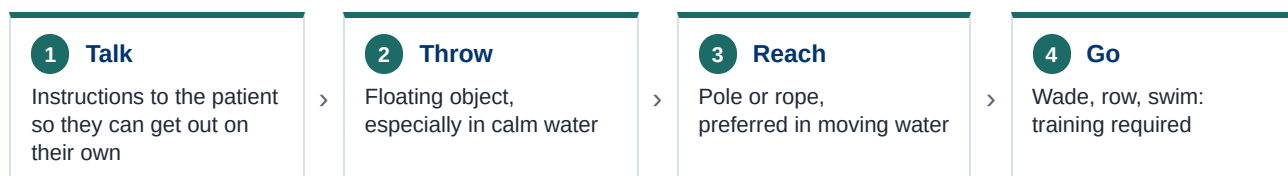


Figure 3. The ladder approach: the first three steps are defensive, the last is offensive.

- 1 Defensive first, offensive never without training.** Talk, throw, reach: what any first aider can do from shore. Wade, row, swim: reserved for trained people, and swimming is a last resort even for them. If you reach the patient: give them something that floats, tow them, carry them.
- 2 Cold water acts in three stages: 1-10-1.** One minute to get your breathing under control (involuntary gasp, hyperventilation, panic, pulse and blood pressure climbing, possible arrhythmias). Ten minutes of useful movement for self-rescue. About one hour before hypothermia causes loss of consciousness.
- 3 Patient out of the water: ABC, CPR, vomiting.** An unconscious patient who is not breathing receives CPR immediately. Expect vomiting; recovery position as soon as breathing resumes. Treat hypothermia, evacuate: respiratory complications can appear hours later.
- 4 What influences survival.** Water temperature (cold protects the brain but causes cold shock and hypothermia; in very cold water, the odds are better), the patient's condition (children and vulnerable people are at higher risk), water quality (lung infections), speed of resuscitation and oxygenation.
- 5 Submersion time guides the decision.** After 30 minutes: reassess resources, risks and benefits. Between 60 and 90 minutes: survival rare but possible (child, very cold water). Beyond that: the operation shifts from rescue to recovery, with minimal risk to rescuers. These decisions belong to advanced water rescue training.